

Metabolomics & Human Health



October 5-9, 2026

Paul Langevin Center CAES CNRS, Aussois, Savoie

**FREE OF
CHARGE**

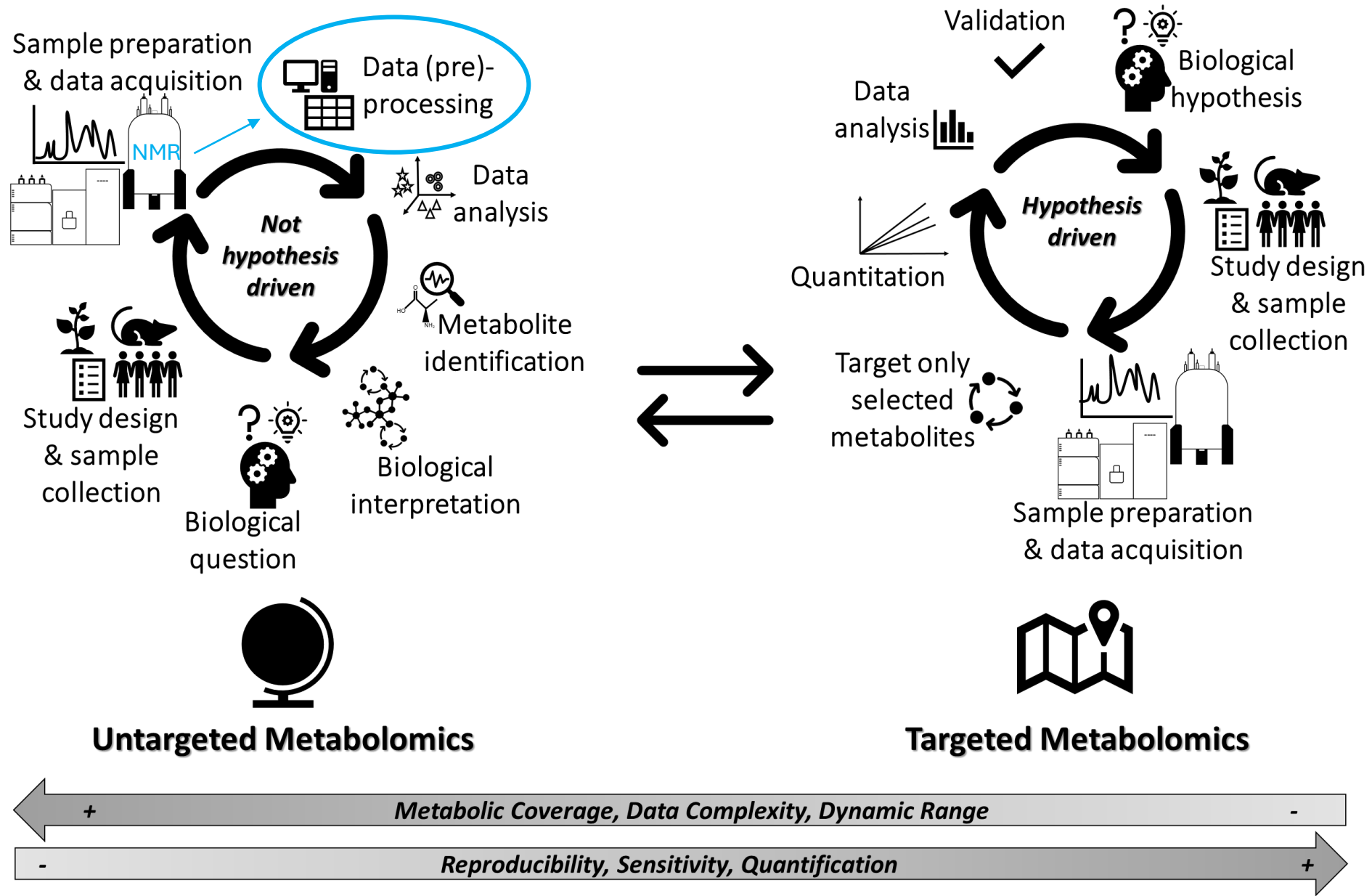
FOR CNRS EMPLOYEES

Session NMR-Wednesday morning

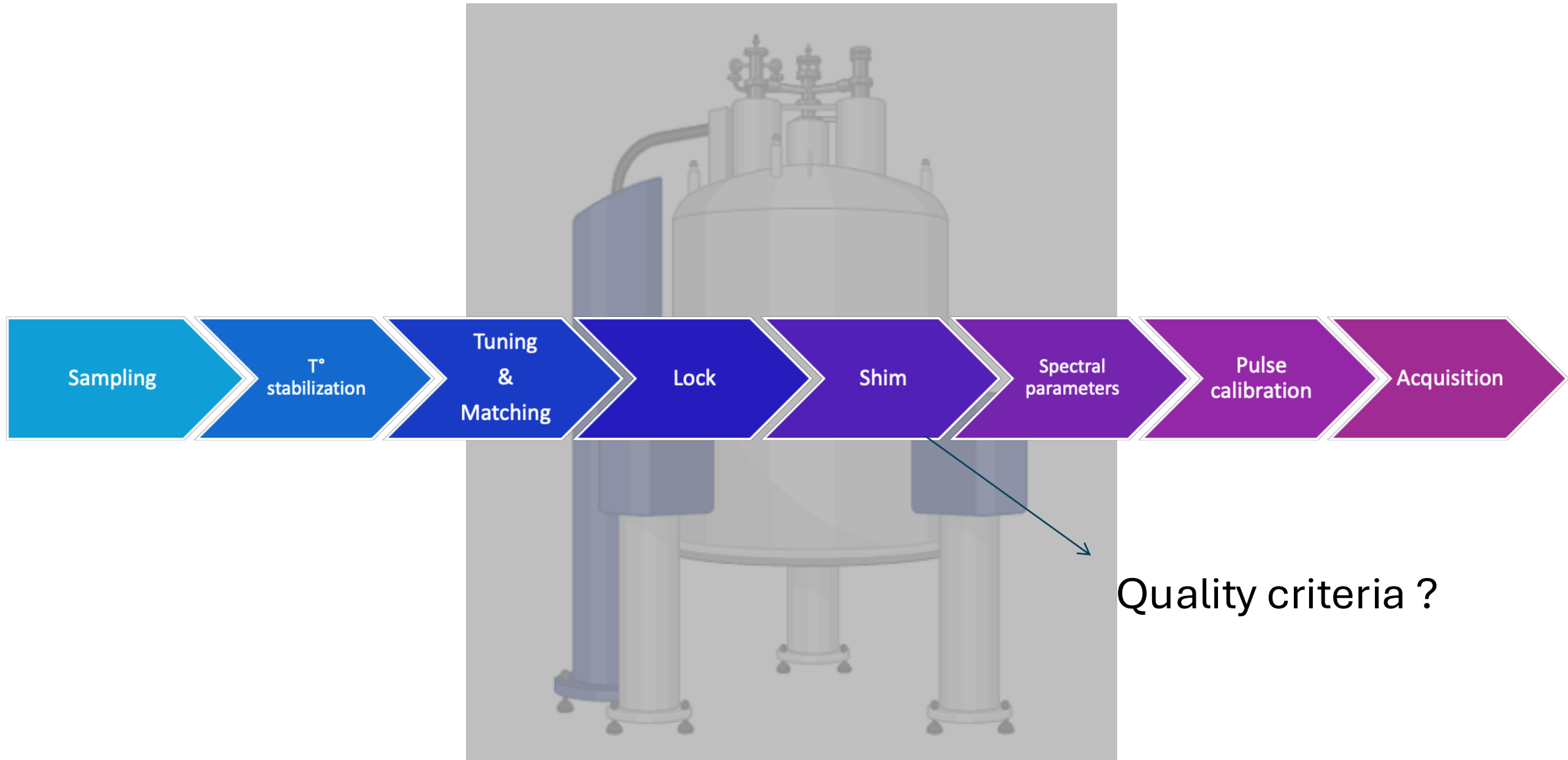
Workshop 2 : NMR data pre/post-processing : from spectra to metabolites

Corentine GOOSSENS, CRIOBE-UPVD, Perpignan

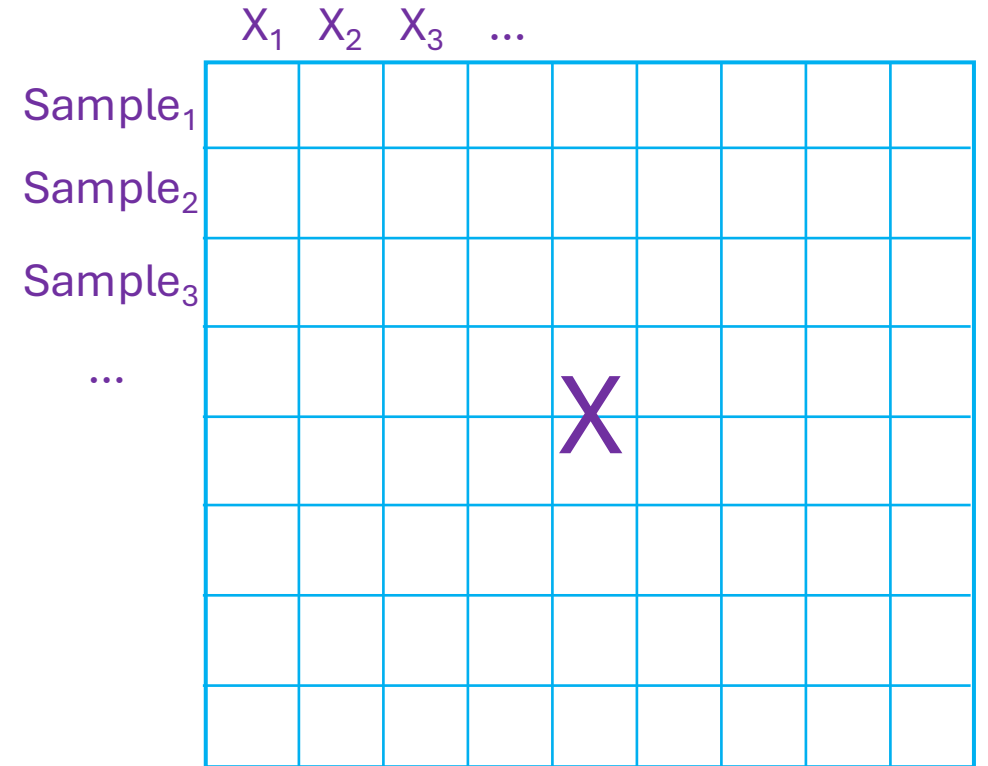
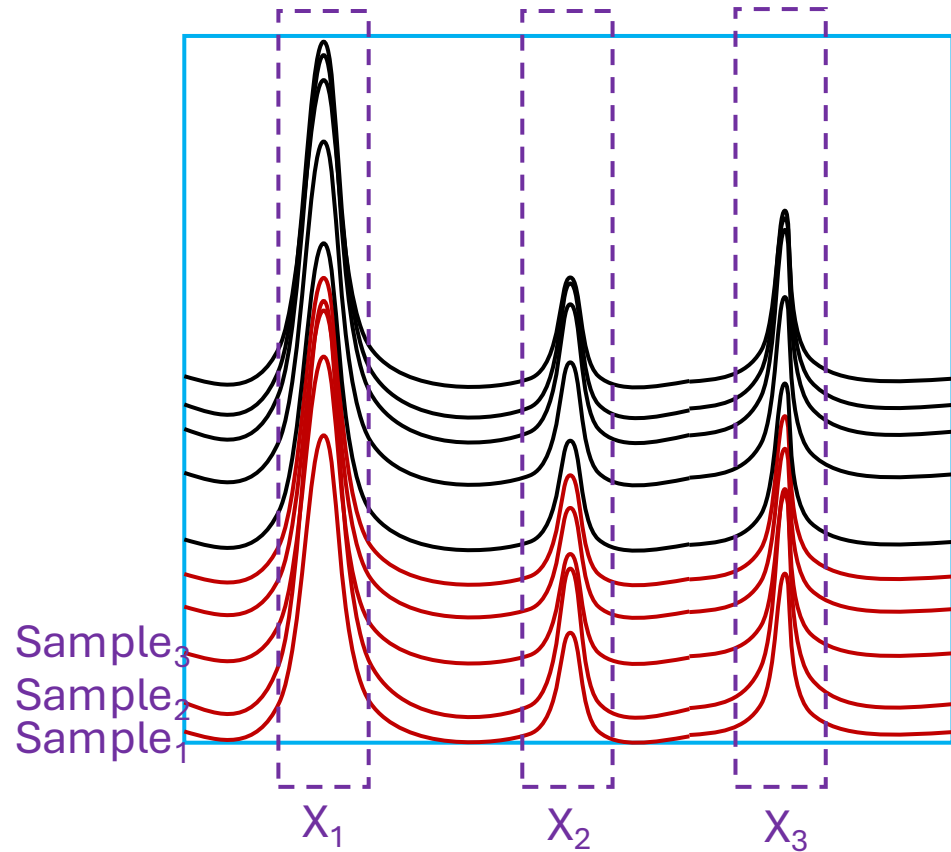
Metabolomics – *Untargeted NMR approach*



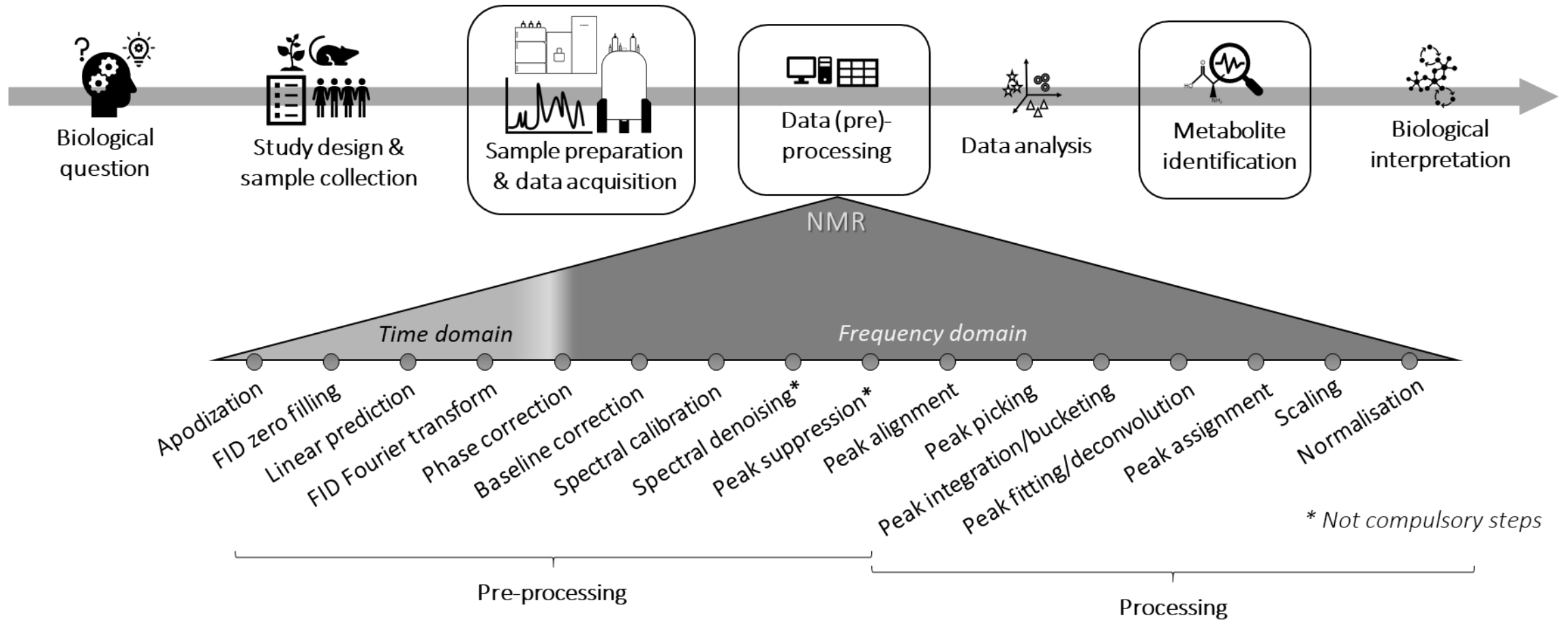
NMR routine & Acquisitions parameters



From raw data to a data matrix



Quite a few steps to process your data



Basic steps, but must be performed very carefully and with high reproducibility !

NMRProcFlow



NMRProcFlow: Spectral processing for 1D NMR - ver. 1.4.16

Load

Processing

An easy graphical tool dedicated to 1D NMR spectra processing for metabolomics

Instrument/Vendor/Format:

Jeol (JDF/DELTA)

Spectra type:

FID

Parameters

ZIP file

Browse...

No file selected

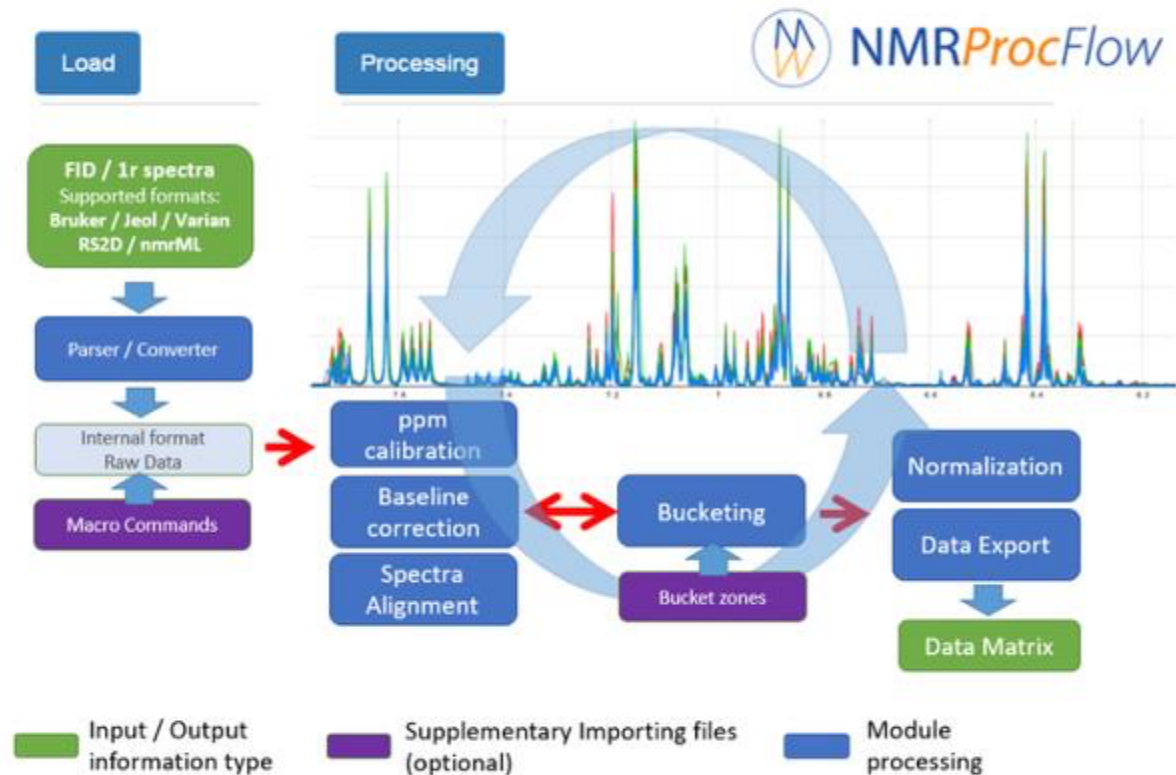
Samples file (Tabular format)

Browse...

No file selected



Get more information on input data format



METABOLOMICS APPROACHES

[General introduction](#)
[Metabolic Fingerprinting](#)
[Targeted Metabolomics](#)
[Processing methods](#)

QUICK TUTORIAL

[1- Data preparation phase](#)
[2- View the spectra](#)
[3 - Interactive data processing](#)
[4 - Spectra Processing](#)
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[6 - Data Export](#)
[7 - Restore a session](#)
[8 - Batch mode execution](#)

EXAMPLES

[Examples in action](#)
[A full data set as example](#)

Data preparation phase

The current version of **NMRProcFlow** accepts raw data come from four major vendors namely Bruker GmbH, Agilent Technologies (Varian), Jeol Ltd and RS2D. Moreover, we also support the nmrML format (See nmrml.org for further information on this format)

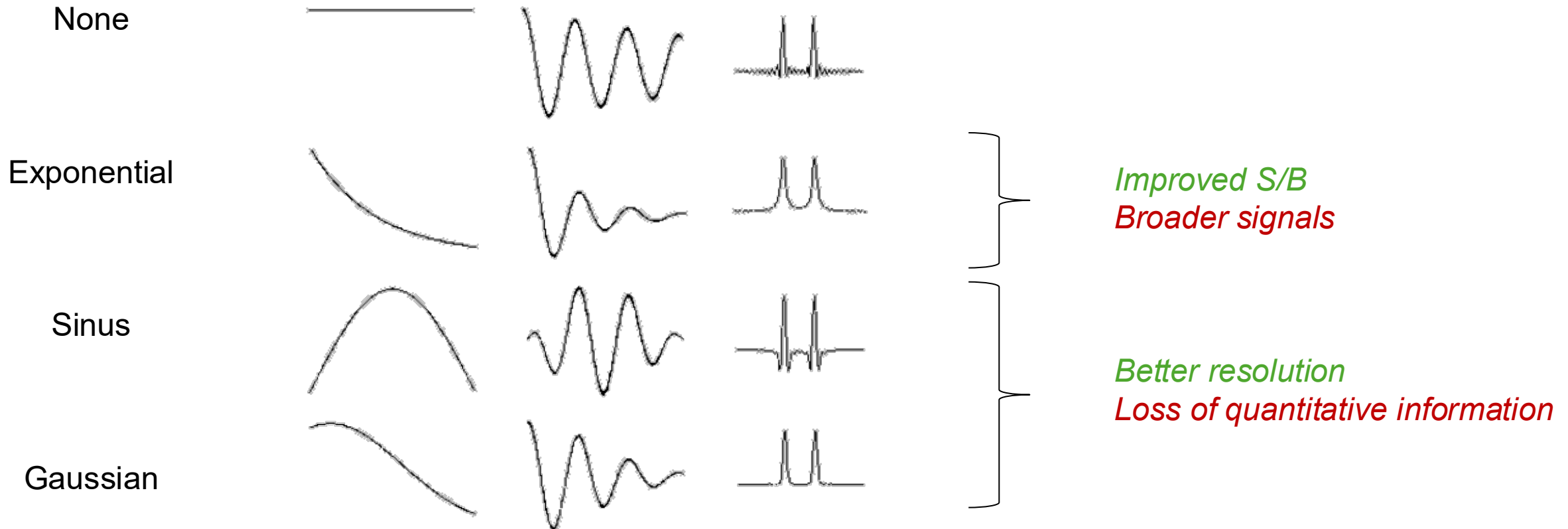
[Bruker](#)[Varian/Agilent](#)[Jeol](#)[RS2D](#)[nmrML](#)

Since March 2018, the NMR Jeol spectra are supported (only the Free Induction Delay) within the JDF format created from the [Jeol Delta software](#).

- See the [Spectral pre-processing for 1D NMR](#) section.
- See a complete example with a JEOL spectra set (JDF Format) - ([PDF online](#))

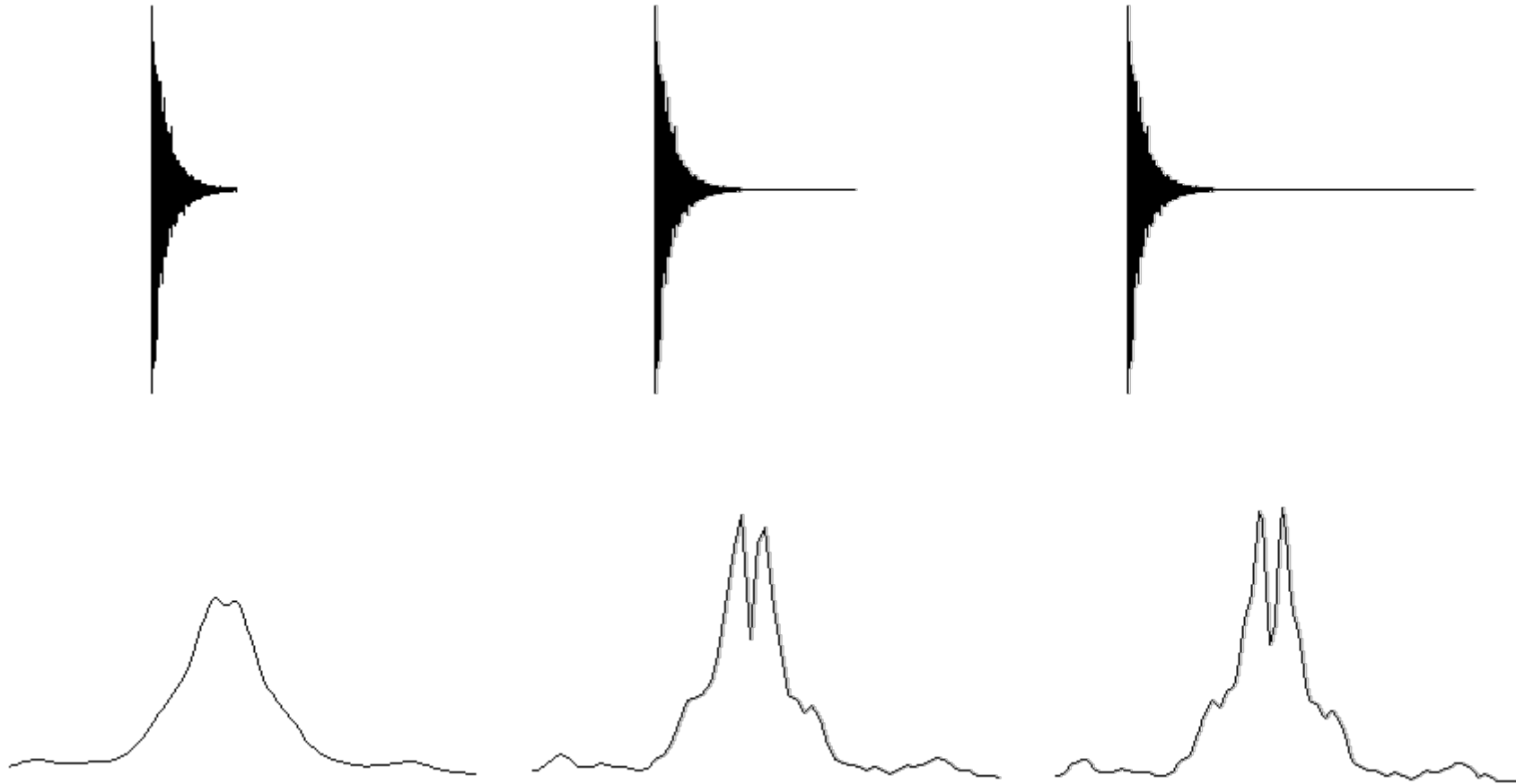
Apodization

Multiply the FID signals by a mathematical function to improve the quality of the frequency domain spectra.



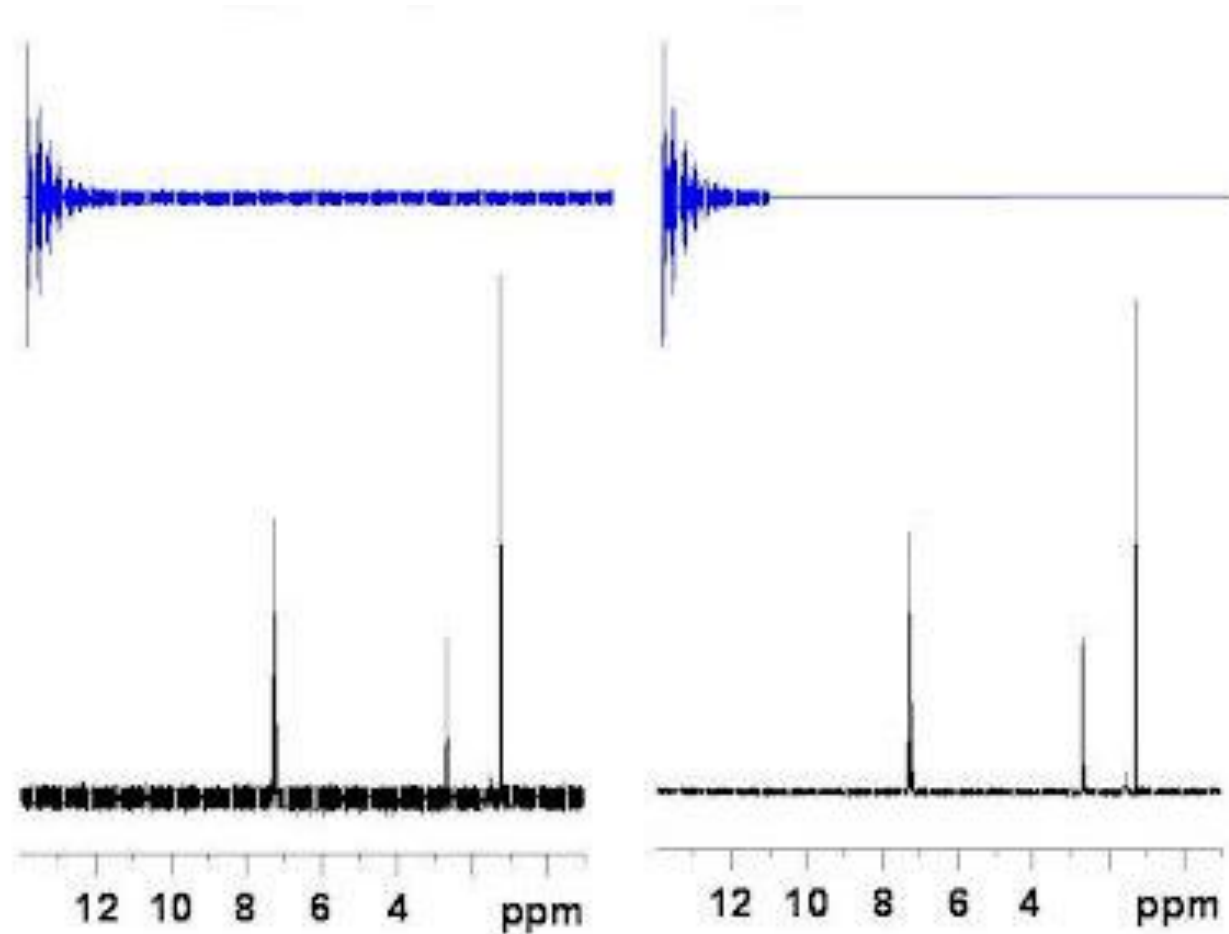
FID zero-filling

Add zero at the end of the FID signal to improve the digital resolution of the frequency domain spectra.



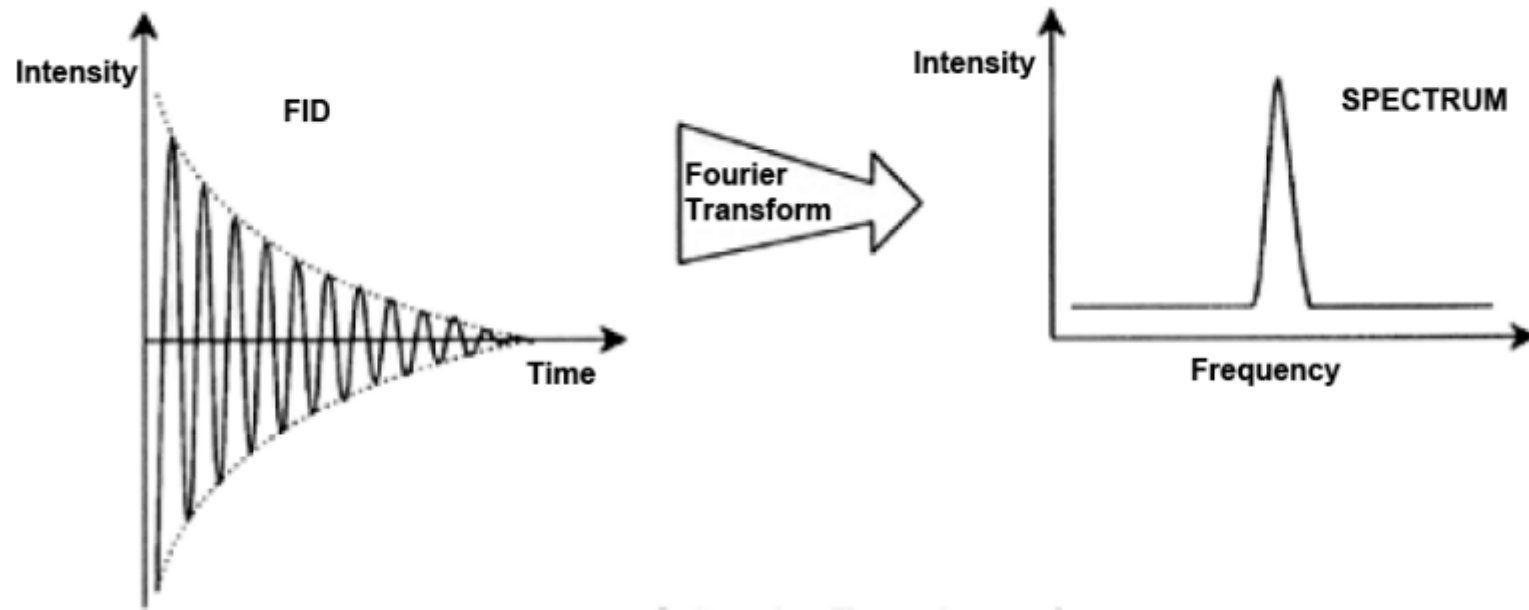
Spectral denoising

Remove noise and improve SNR.



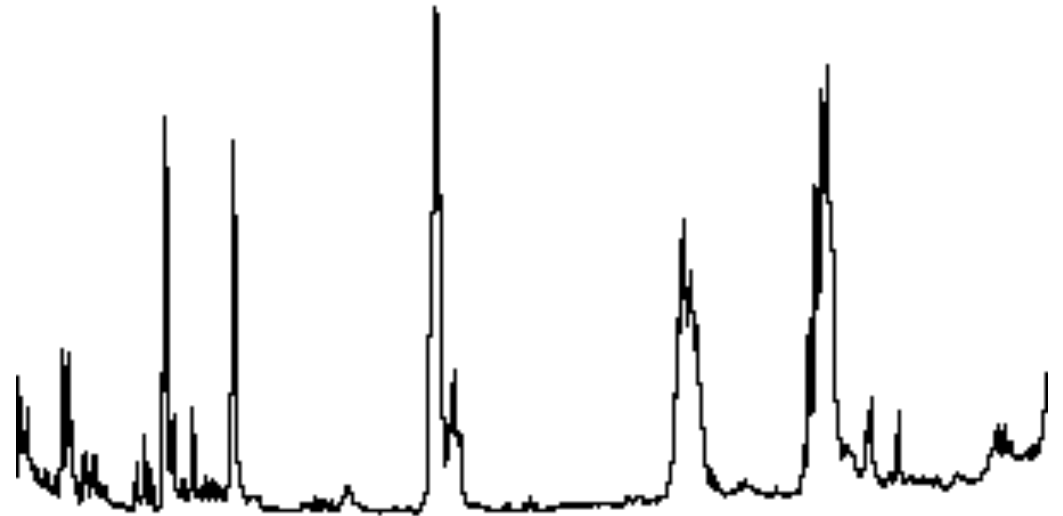
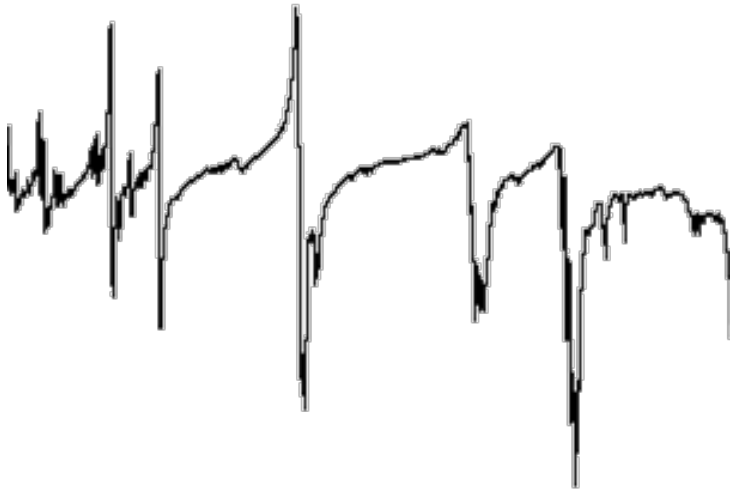
FID Fourier Transform

Convert the Free Induction Decay which is in time domain into a spectra in frequency domain.



Phase correction

Correct the phase shift induced by the gap between the detection coil and the transverse magnetization at the beginning of the FID signals.



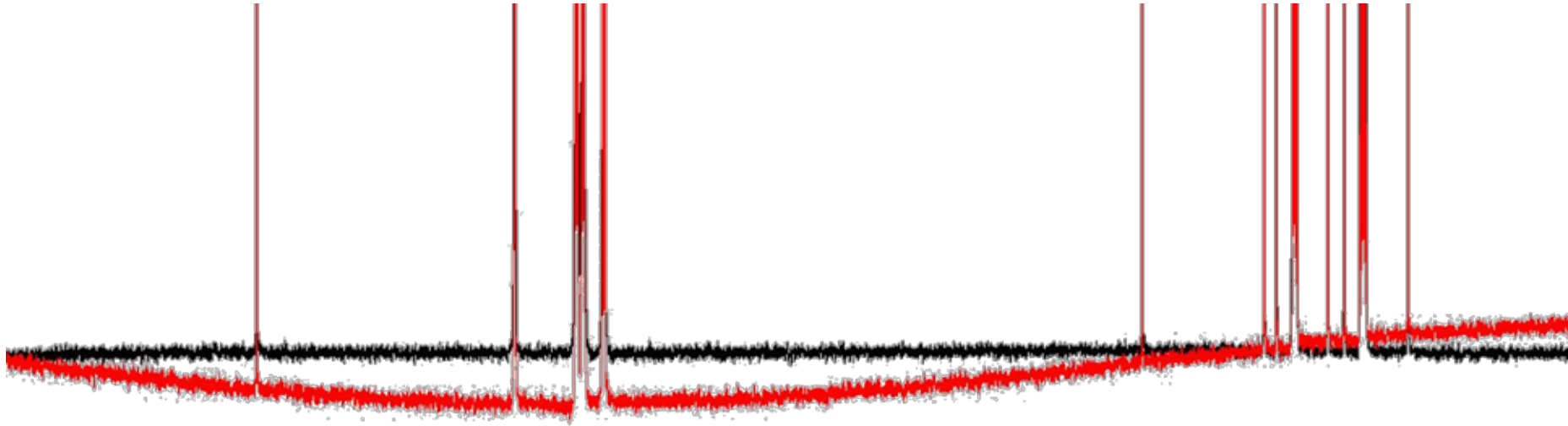
Automatic (faster) or manual (cleaner)

-> 0 order

-> (1st order)

Baseline correction

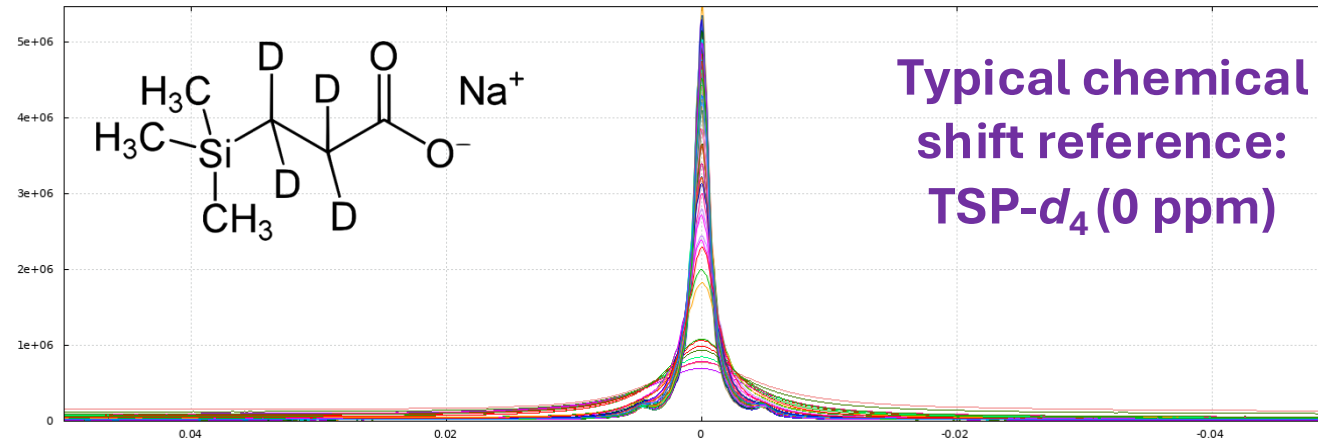
Correct the distortion of the baseline and ensure that the integration of the signals is properly done.



Automatic (faster) or manual (cleaner)

Spectral calibration: the importance of referencing

Calibrate all the spectra of a cohort to the same chemical shift referencing to facilitate alignment, bucketing and others processing steps.

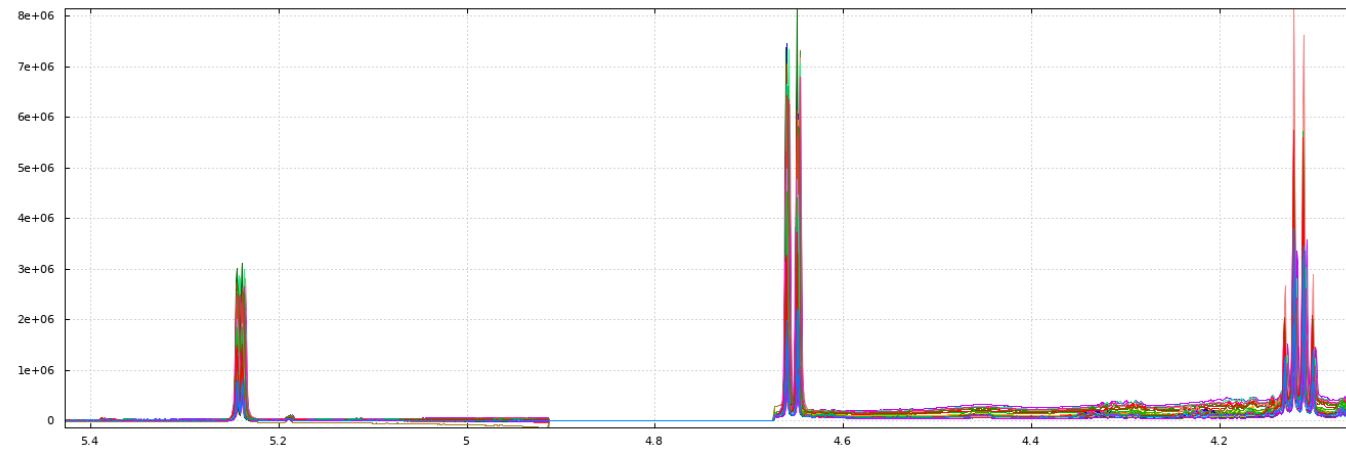
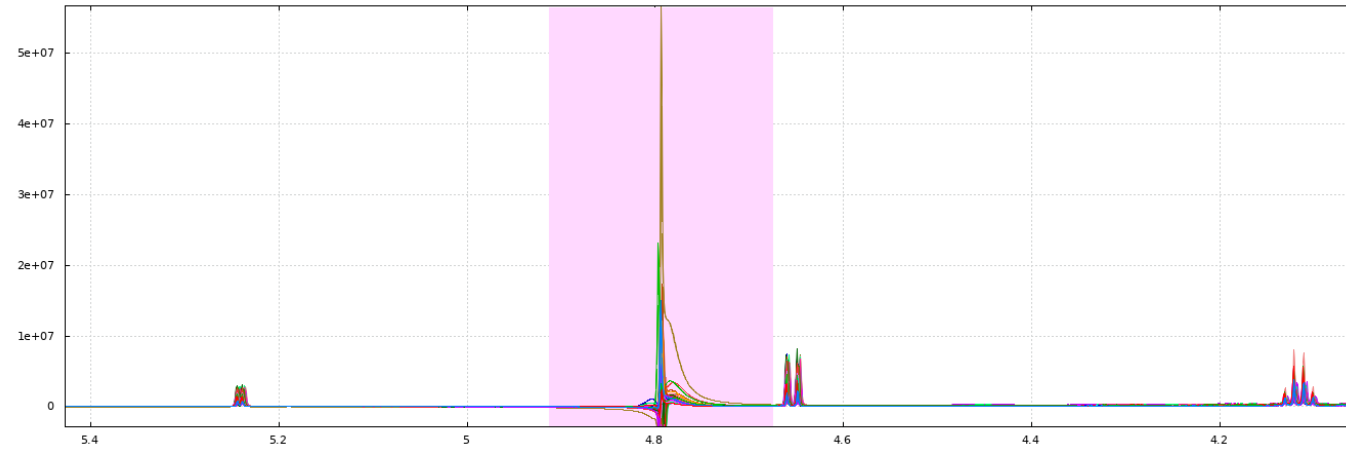


Criteria for choosing a reference:

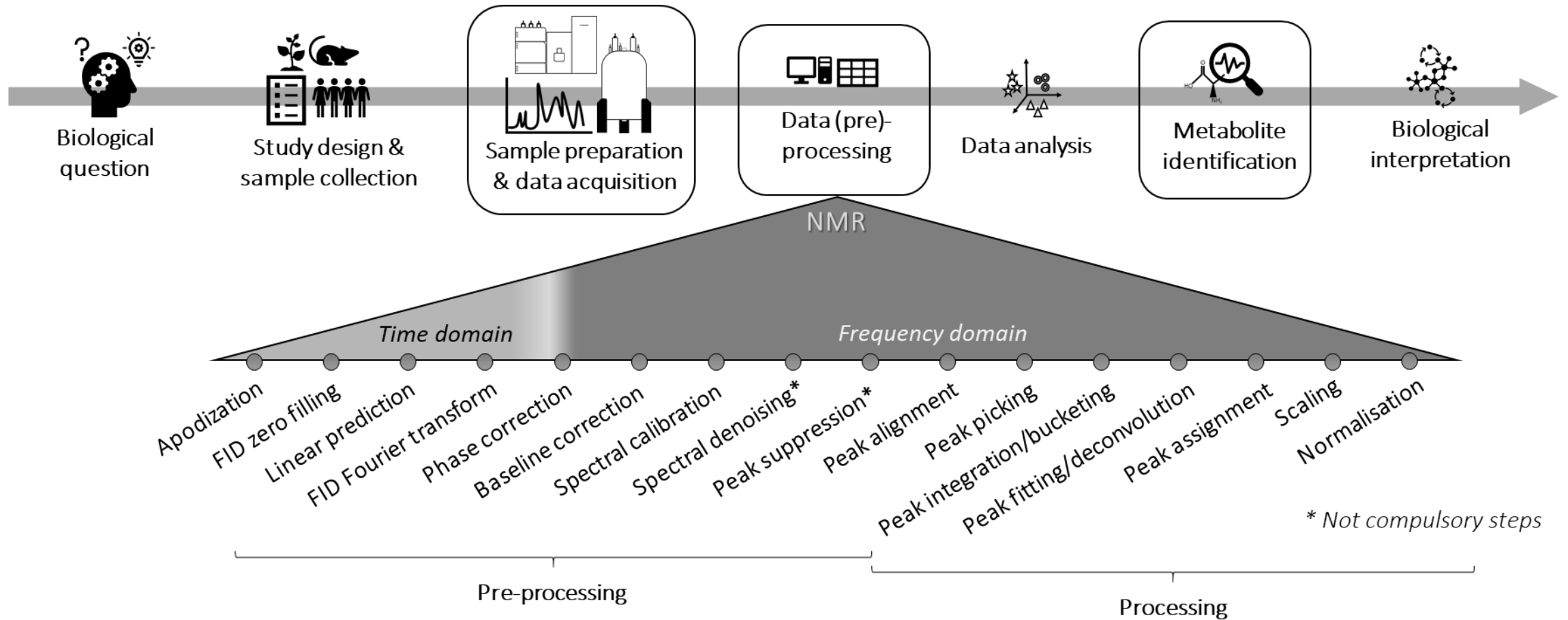
- Soluble within the sample
- No interaction with the sample
- Distinct chemical shift

Peak suppression

Remove unwanted signals without any biological information (solvent, urea,...).



Quite a few steps to process your data



Basic steps, but must be performed very carefully and with high reproducibility !

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QUICK TUTORIAL

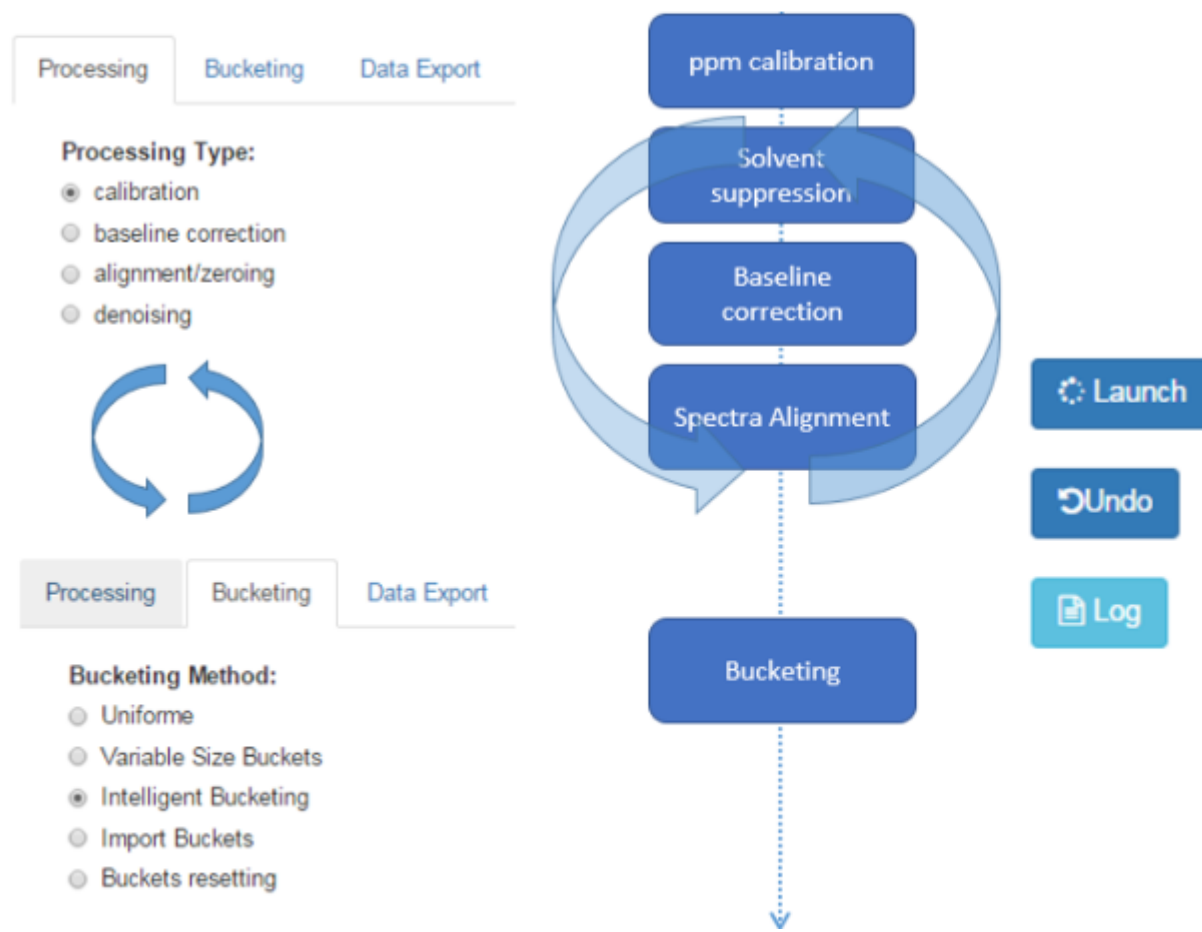
[1- Data preparation phase](#)[2- View the spectra](#)[3 - Interactive data processing](#)[4 - Spectra Processing](#)[5 - Bucketing](#)[6 - Data Export](#)[7 - Restore a session](#)[8 - Batch mode execution](#)

EXAMPLES

[Examples in action](#)[A full data set as example](#)

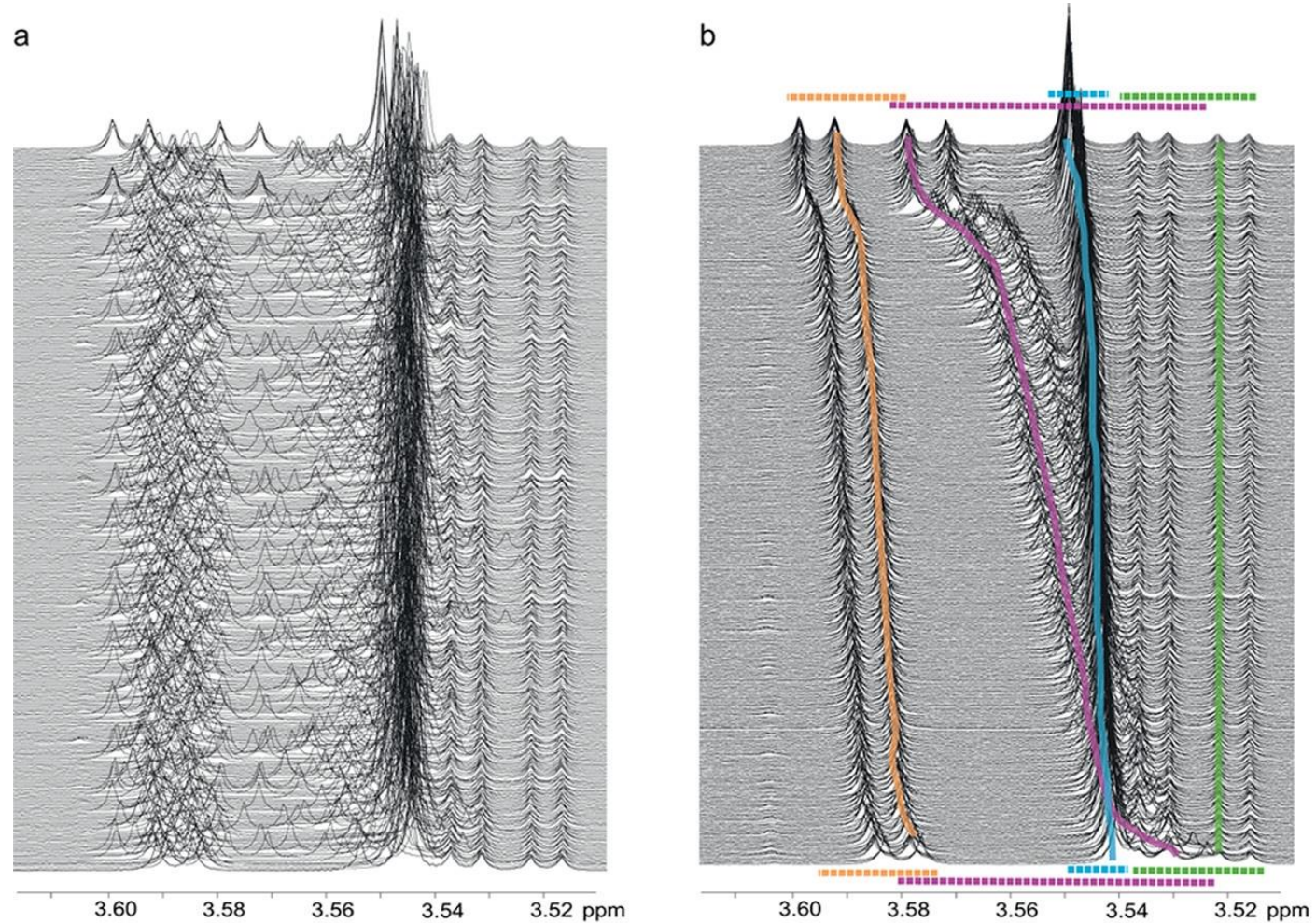
Interactive data processing

It is possible to navigate between tabs and then **launch any processing in the order you want**.



Signal alignment

Correct slight changes due to pH, ionic strength, temperature, protein content...

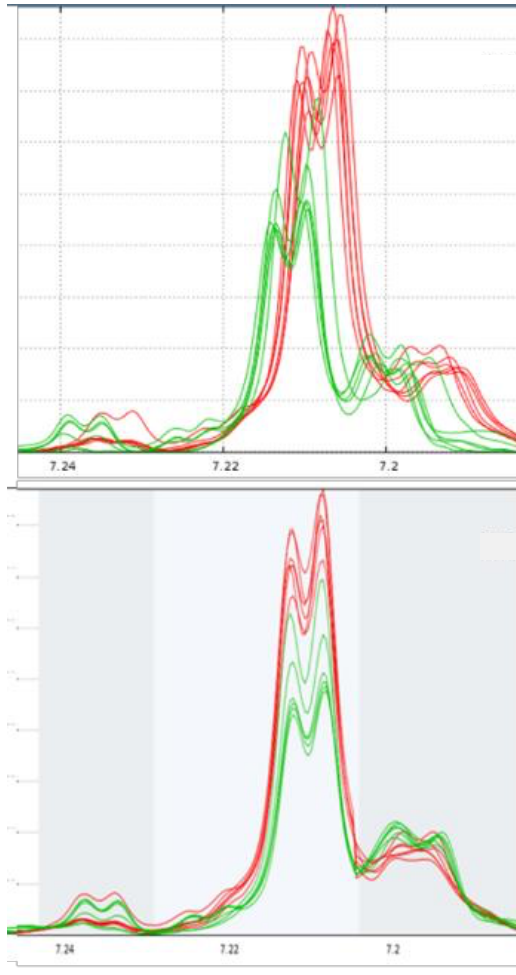


Signal alignment

Use specific algorithms to globally or locally correct misalignment.

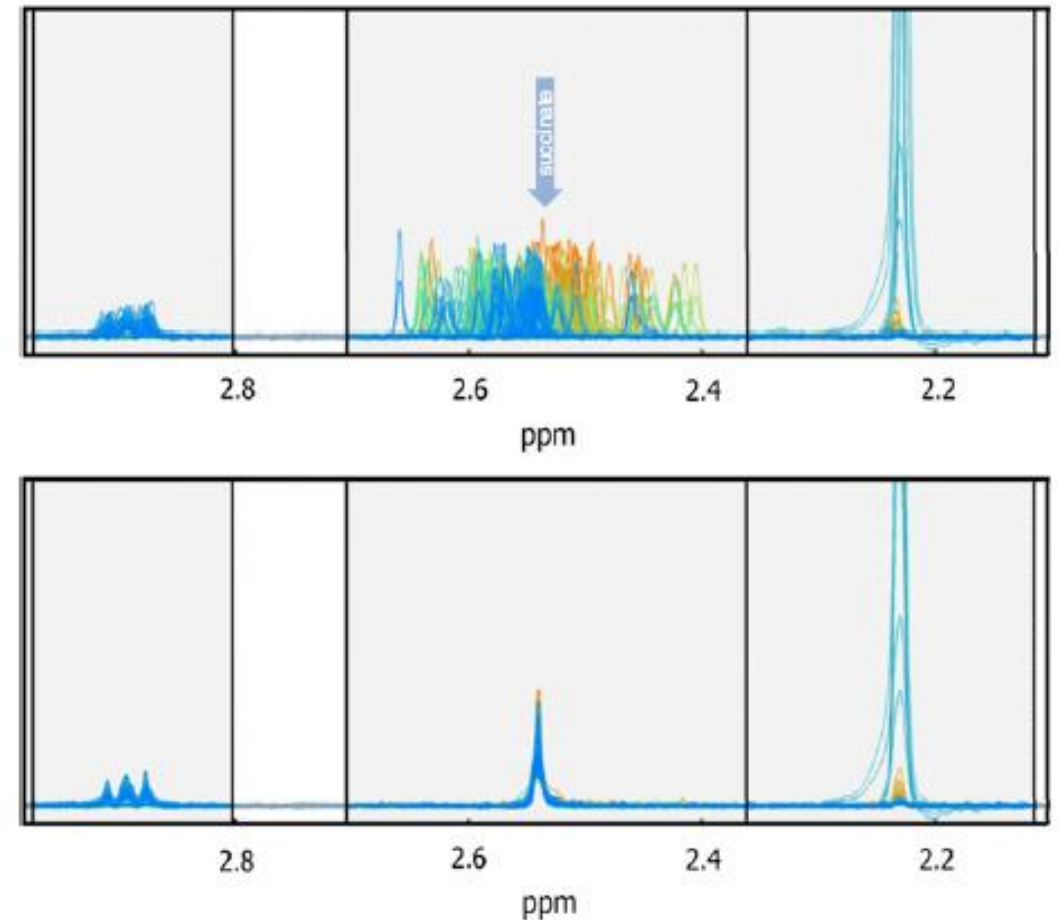
<https://nmrprocflow.org/>

**Before
alignment**

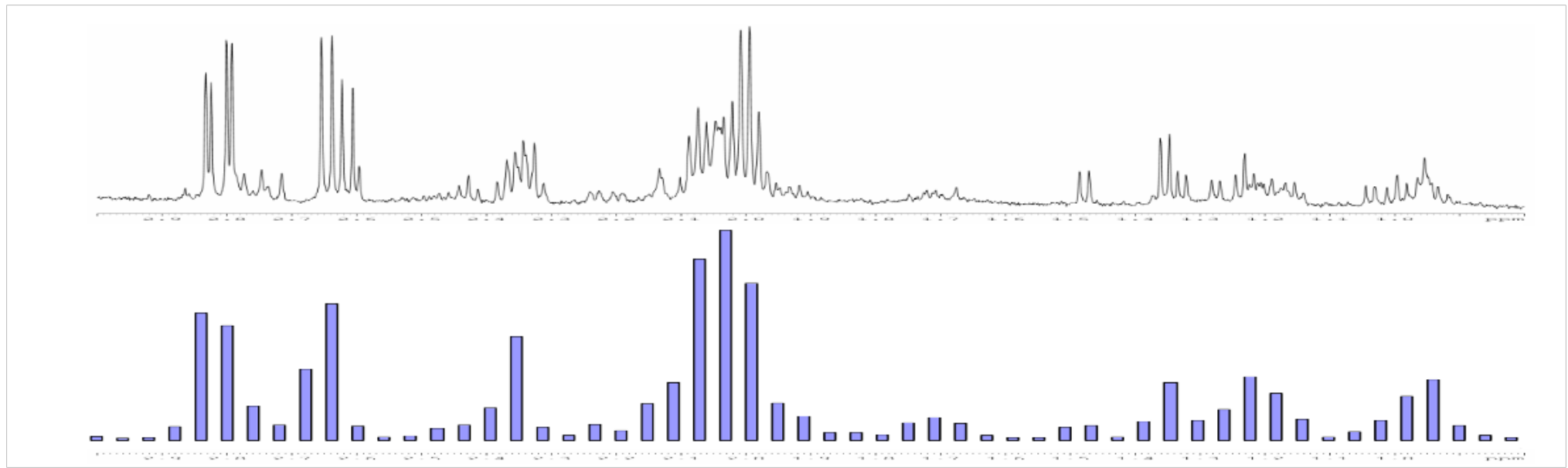
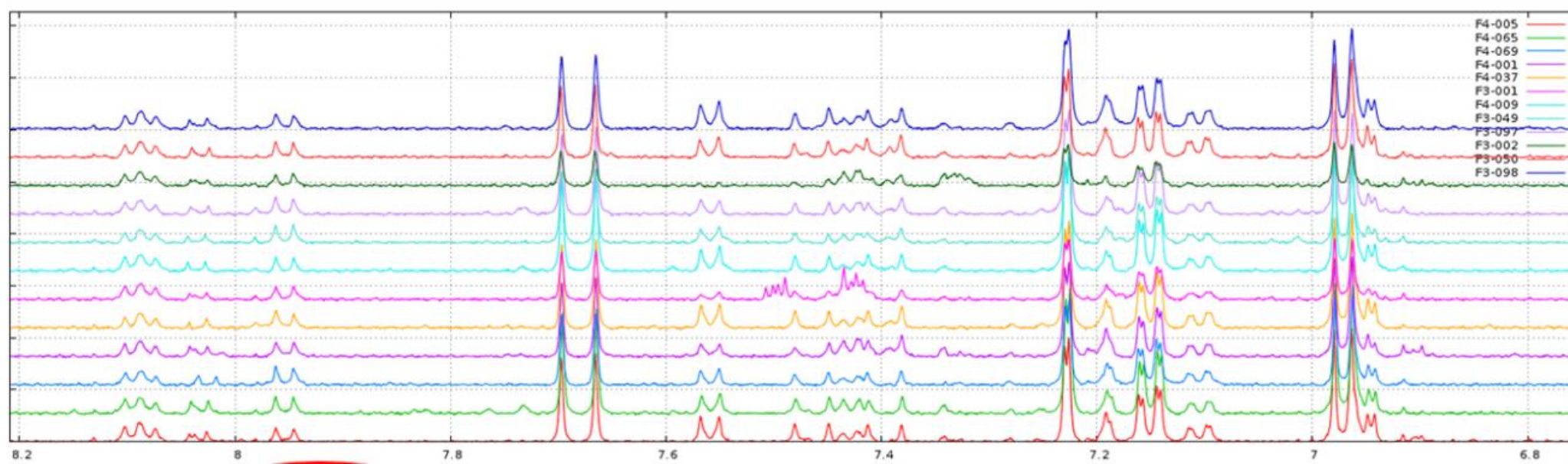


**After
alignment**

<https://github.com/mfitzp/icoshift>

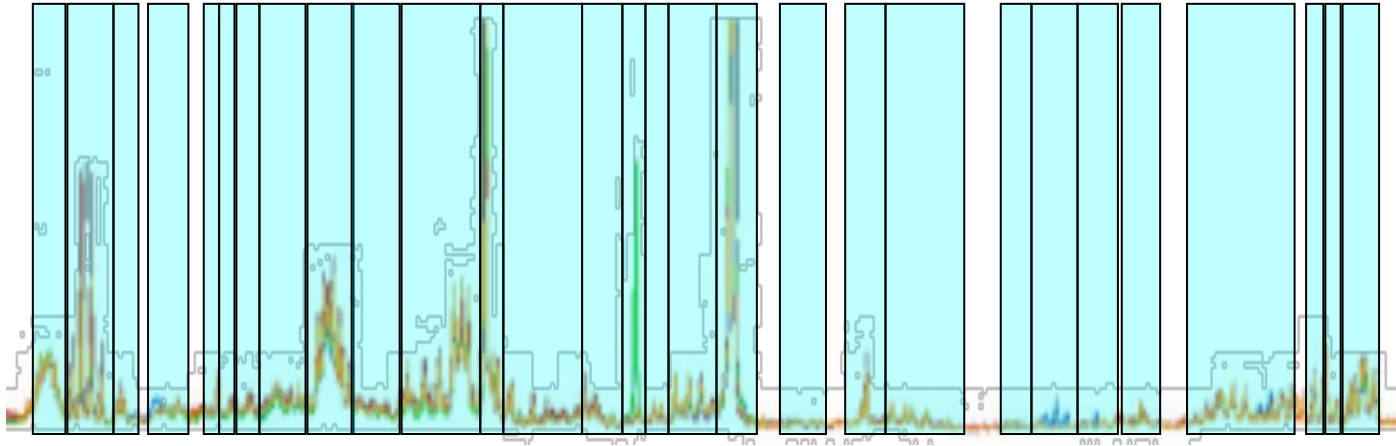


Spectral bucketing / binning



Spectral bucketing / binning

Divide the spectra into fixed or variable sized buckets for data reduction and integration prior statistical analysis.



All spectra are divided into small regions for data reduction prior to statistical analysis;

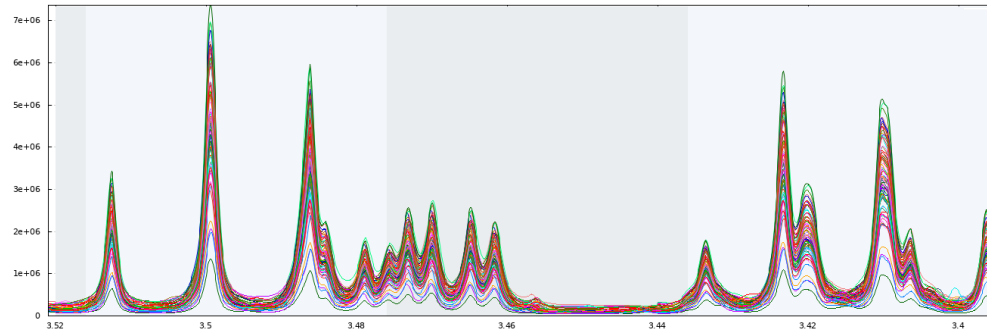
Binning (or bucketing) can be performed with constant (eg. 0.04 ppm) or variable width;

Intelligent bucketing methods should be preferred to avoid arbitrary data splitting;

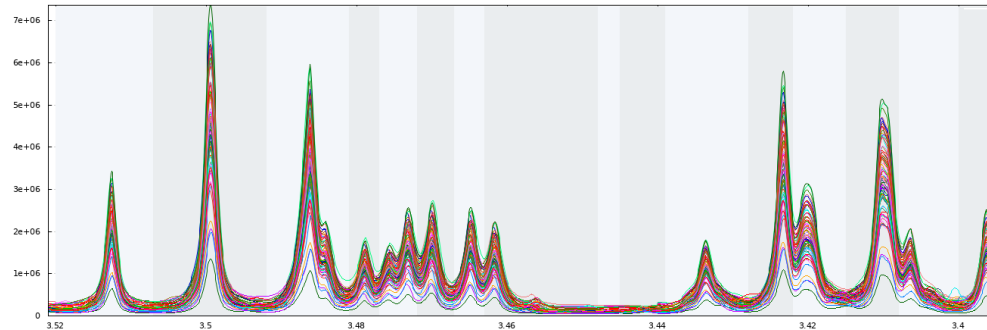
Dedicated free software exist (MVAPack, W4M, NMRProcFlow, Chenomx...)

Spectral bucketing / binning

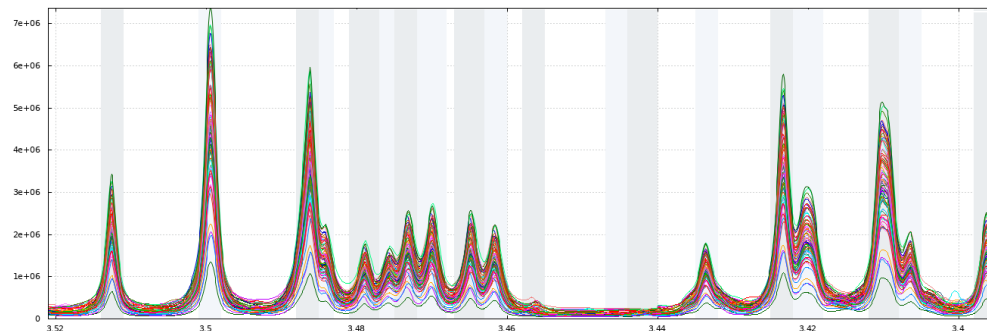
Divide the spectra into fixed or variable sized buckets for data reduction and integration prior statistical analysis.



Fixed size buckets
(0.04 ppm)



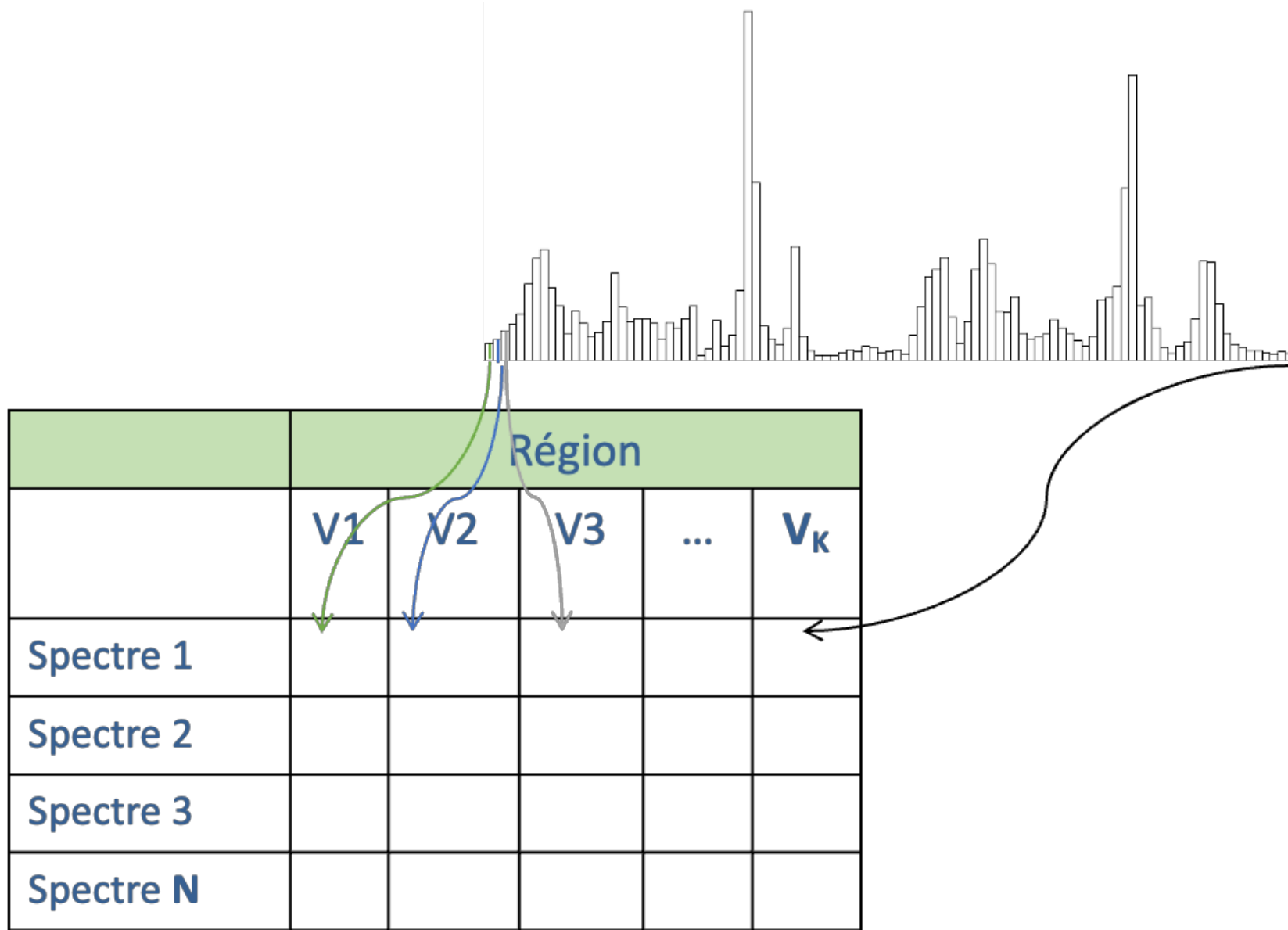
Variable size buckets thanks to the
Intelligent bucketing algorithm



Variable size buckets thanks to the
ERVA bucketing algorithm

Peak integration

Integrate the area under the curve for each buckets to obtain a data matrix.



Data matrix

Processing

Bucketing

Data Export

Data Type to Export:

☒ Data matrix

☐ Buckets table

☐ SNR matrix

☐ XLSX Workbook

☐ Spectral data

Export Format:

Comma Separator Value (CSV)

SNR threshold:

3

noisy PPM range:

10.5 10.2

Normalization Method:

Constant Sum Normalization

PPM range of the Reference:

Export Data Matrix

After exporting, the data matrix is formatted so that we can subsequently perform statistical analysis using BioStatFlow web application. Thus the data file manipulations are minimized. See [Metabolic Fingerprinting](#)



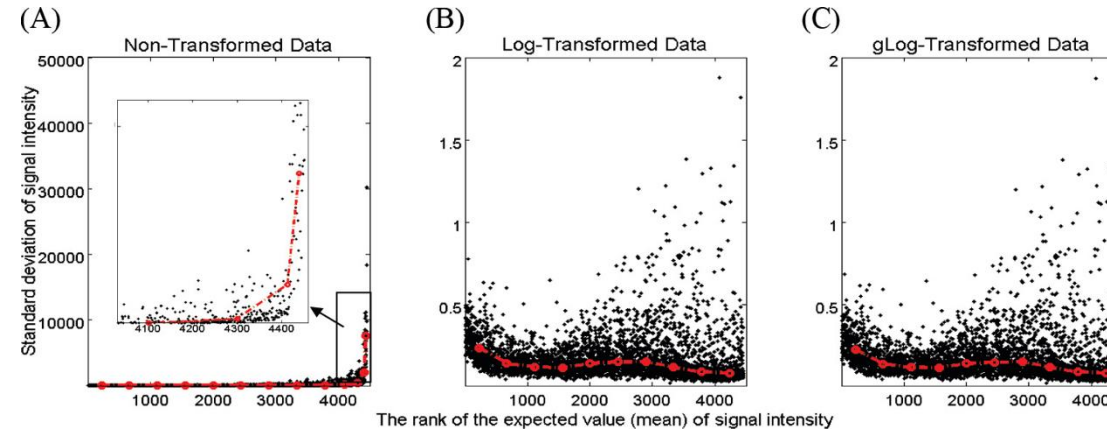
Export Data Matrix

Samplecode	Condition	Stage	B9_1272	B8_5408	B8_4573	B8_2825	B7_6956	B7_6635	B7_4512	B7_4362	B7_4224	B7_4090	B7_3957	B7_3834	B7_3711
F3-001	Control	J08	0.11928791	0.00581534	0.16160283	0.047086	0.09814824	0.09813412	0.01898936	0.03941068	0.04948976	0.02022299	0.00373654	0.02378087	0.02315004
F3-049	Control	J08	0.10384242	0.00835102	0.13589457	0.06159699	0.14618664	0.14036055	0.01539754	0.01692683	0.01200164	0.01717897	0.01085015	0.02315004	0.02315004
F3-097	Control	J08	0.09651629	0.00623615	0.17493857	0.06213409	0.16545419	0.18503065	0.02161687	0.0214187	0.02813971	0.02358135	0.01270131	0.02834738	0.02834738
F3-002	Shadow	J08	0.09611617	0.0045809	0.17093983	0.04593351	0.0781413	0.06906432	0.01626121	0.02980209	0.03109361	0.01965519	0.01404522	0.02284019	0.02284019
F3-050	Shadow	J08	0.12598911	0.01304445	0.13824284	0.05809637	0.14435492	0.13533937	0.03232633	0.01034411	0.01846426	0.02983426	0.02225713	0.03401898	0.03401898
F3-098	Shadow	J08	0.12360064	0.00724083	0.12727933	0.05951439	0.17228824	0.16153312	0.04384093	0.01533951	0.01755694	0.03644395	0.01806933	0.03015551	0.03015551
F3-013	Control	J15	0.05330992	0.00196927	0.13305672	0.0445001	0.0490723	0.04415489	0.01263968	0.03341843	0.04544959	0.02119349	0.01533574	0.02041401	0.02041401
F3-061	Control	J15	0.06667186	0.00552217	0.08759005	0.04733081	0.03781117	0.03933464	0.02078347	0.05830485	0.07758045	0.0348291	0.01933022	0.03107577	0.03107577
F3-109	Control	J15	0.07548592	0.0043453	0.08763526	0.04697576	0.05321745	0.05536447	0.00768144	0.0600623	0.08110839	0.03416974	0.01394663	0.03504505	0.03504505
F3-062	Shadow	J15	0.05708147	0.00229657	0.07625981	0.04708255	0.03478851	0.02437456	0.01564896	0.04840928	0.0614291	0.02940266	0.01571519	0.02557809	0.02557809
F3-110	Shadow	J15	0.08843008	0.00366854	0.11023306	0.04240016	0.04629357	0.0565123	0.02686256	0.03765864	0.046656	0.02004858	0.01410488	0.02813436	0.02813436
F3-025	Control	J15	0.04971968	0.00192048	0.07801866	0.02777739	0.05304738	0.02581476	0.00771899	0.05555394	0.06138559	0.03070073	0.01540366	0.02825725	0.02825725
F3-073	Control	J15	0.05267122	0.0042229	0.09108597	0.03208542	0.04221378	0.02215247	0.02931834	0.07019428	0.0875616	0.04830502	0.02846465	0.04251691	0.04251691
F3-121	Control	J28	0.04971739	0.00247732	0.06571802	0.04087093	0.01940916	0.01839539	0.01829224	0.08595428	0.11116443	0.04523421	0.02985821	0.04415303	0.04415303
F3-026	Shadow	J28	0.0418849	0.00111314	0.06341996	0.02779082	0.01097475	0.01296508	0.01251698	0.0412173	0.04200217	0.0241513	0.01693586	0.02228252	0.02228252
F3-074	Shadow	J28	0.05363533	0.00261704	0.07941371	0.02939763	0.0131005	0.02418532	0.01724451	0.08383088	0.10181302	0.04628116	0.02906383	0.04369965	0.04369965
F3-122	Shadow	J28	0.05738645	0.00203373	0.06196753	0.04248931	0.02371078	0.01953007	0.03795215	0.08552814	0.10944816	0.04647477	0.03542677	0.04118655	0.04118655
F3-037	Control	J55	0.05166236	0.04100089	0.05889308	0.09333538	0.0081629	0.00884594	0.03447293	0.07293427	0.08325465	0.03443659	0.02862395	0.03033902	0.03033902
F3-085	Control	J55	0.06712416	0.06835656	0.10920451	0.11024554	0.01595192	0.00922174	0.028818	0.07459614	0.07824352	0.02872809	0.04676672	0.03166962	0.03166962
F3-133	Control	J55	0.04319144	0.08858712	0.121372	0.13439019	0.00701626	0.01108497	0.02752508	0.09864951	0.11831952	0.04616909	0.05693713	0.04737119	0.04737119
F3-038	Shadow	J55	0.07895642	0.12345603	0.1278891	0.1784346	0.00541609	0.0076085	0.03272434	0.10715168	0.12007041	0.04038197	0.05594181	0.04416299	0.04416299
F3-086	Shadow	J55	0.07690517	0.09031835	0.12899566	0.14985144	0.01430116	0.00886306	0.01626272	0.06070653	0.07374108	0.02860104	0.05225686	0.02533339	0.02533339
F3-134	Shadow	J55	0.05143253	0.1055738	0.12400084	0.14763796	0.01646724	0.00440921	0.03385417	0.11258537	0.12189743	0.04185418	0.04906765	0.04950948	0.04950948
F4-001	Control	J08	0.1230095	0.0089157	0.16097331	0.05711378	0.14520676	0.16760372	0.03105041	0.03127852	0.03201556	0.02952918	0.02039009	0.0337445	0.0337445

Note that the factors are embedded in the file, provided they have been specified in the first step (file samples)

Centring / Scaling

Compensate for the large dynamic range and the influence that it can have on statistical analyses.



UPLC-MS data (urine)

Veselkov et al., *Anal. Chem.*, 2011

I. Centering
Mean-centering:
 $x \rightarrow x - \text{mean}(x)$

II. Scaling
Unit variance: $x \rightarrow x/\sigma$
Pareto: $x \rightarrow x/\sqrt{\sigma}$
Range : $x \rightarrow x/|x_{\max} - x_{\min}|$
Level: $x \rightarrow x/\text{mean}(x)$

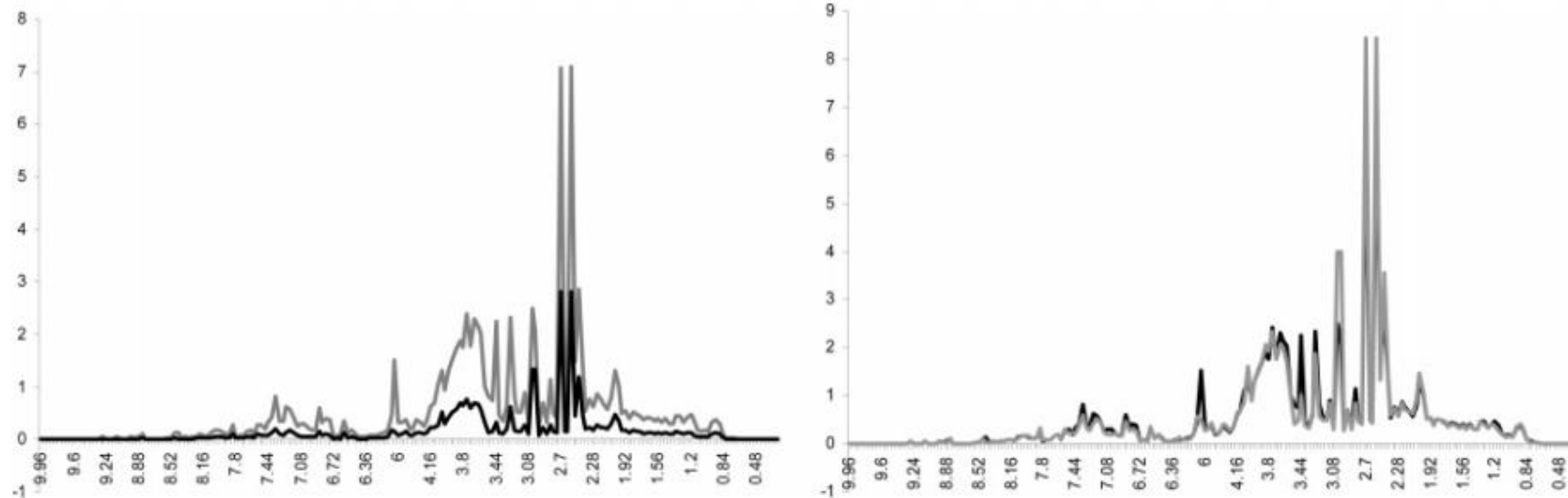
III. Transformation
Log:
 $x \rightarrow \log(x)$

Van Den Berg et al., *BMC Genomics*, 2006

Normalization

Overcome dilution problems (especially for urine samples), pipeting errors or analytical variations to allow multiple spectra comparison.

Numerous methods exist: total sum normalization, probabilistic quotient normalization, normalization to an internal reference... Each of them having their drawbacks and strengths.



Summary of the (pre)-processing steps

